

① 문제만 보고 가로 빈칸에 직접 풀이 ② 막히면 풀이 폴더의 같은 번호 .md 로 대조 ③ 채점: solve-exam-prep · 정리노트 1순위

떠올릴 개념

Ch9 · 정현파 · phasor

- $v = V_m \cos(\omega t + \phi)$, $\omega = 2\pi f$, $T = 1/f$
- $Z_R = R$, $Z_L = j\omega L$, $Z_C = 1/(j\omega C)$
- 페이지저 $V = Z \cdot I$ — 전압분배 · 테브난 그대로 적용

Ch10 · 최대전력전달

- $Z_L = Z_{Th}^*$ (켈레 정합)
- $P_{max} = |V_{Th,rms}|^2 / (4 \cdot R_{Th})$
- 순저항 부하 제약: $R = |Z_{Th}|$

Ch14 · 필터 · 공진

- ω_c : -3dB, $|H| = 1/\sqrt{2}$
- RL LPF $\omega_c = R/L$, RC HPF $\omega_c = 1/RC$
- 공진 $\omega_0 = 1/\sqrt{LC}$, $Q = \omega_0 L/R$, $\beta = \omega_0/Q$

문항 목차 (총 15문항)

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9.1 A sinusoidal current is given by the expression

$$i = 100 \cos (600t + 45^\circ) \text{ mA}.$$

Find (a) f in hertz; (b) T in milliseconds; (c) I_m ; (d) $i(0)$; (e) ϕ in degrees and radians; (f) the smallest positive value of t at which $i = 0$; and (g) the smallest positive value of t at which $di/dt = 0$.

— 풀이 —

9.3 Consider the sinusoidal voltage

$$v(t) = 25 \cos(400\pi t + 60^\circ) \text{ V}.$$

- a) What is the maximum amplitude of the voltage?
- b) What is the frequency in hertz?
- c) What is the frequency in radians per second?
- d) What is the phase angle in radians?
- e) What is the phase angle in degrees?
- f) What is the period in milliseconds?
- g) What is the first time after $t = 0$ that $v = 0 \text{ V}$?
- h) The sinusoidal function is shifted $5/6 \text{ ms}$ to the right along the time axis. What is the expression for $v(t)$?
- i) What is the minimum number of milliseconds that the function must be shifted to the left if the expression for $v(t)$ is $25 \sin 400\pi t \text{ V}$?

9.5 At $t = 5$ ms, a sinusoidal current is known to be zero and going negative. The current is next zero at $t = 25$ ms. It is also known that the current is 50 mA at $t = 0$.

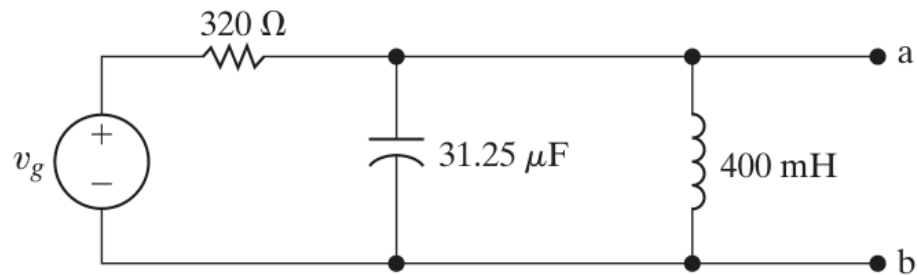
- a) What is the frequency of i in hertz?
- b) What is the expression for i ?

— 풀이 —

9.43 The sinusoidal voltage source in the circuit in Fig. P9.43 is developing a voltage equal to $50 \sin 400t$ V.

- Find the Thévenin voltage with respect to the terminals a,b.
- Find the Thévenin impedance with respect to the terminals a,b.
- Draw the Thévenin equivalent.

Figure P9.43



— 풀이 —

[생성 9.91] 직렬 RLC 회로 — 임피던스 크기·위상

60 Hz 전원 $v(t) = 170 \cos(2\pi \cdot 60 \cdot t)$ [V] 가 $R = 50 \, \Omega$, $L = 0.2 \, \text{H}$, $C = 40 \, \mu\text{F}$ 인 직렬 RLC 회로에 인가된다.

- (a) 리액턴스 $X_L = \omega L$, $X_C = 1/(\omega C)$ 를 구하라.
- (b) 총 임피던스 $Z = R + j(X_L - X_C)$ 와 크기 $|Z| = \sqrt{R^2 + X^2}$, 위상 $\theta = \arctan(X/R)$ 를 구하라.
- (c) 전류 페이저 I (피크와 rms) 를 구하라.
- (d) V_R , V_L , V_C 를 구하고 $E = \sqrt{V_R^2 + (V_L - V_C)^2}$ 로 KVL(페이저 합)을 검증하라.

[생성 문제 · 9~15주차 강의노트 범위 · 해답: 생성문제_해답.md]

[생성 9.92] 직·병렬 회로 — 등가 임피던스

$\omega = 1000 \text{ rad/s}$. 직렬 가지 Z_1 은 $R_1 = 30 \, \Omega$ 과 $L = 40 \text{ mH}$ 의 직렬, 병렬 가지 Z_2 는 $R_2 = 100 \, \Omega$ 과 $C = 10 \, \mu\text{F}$ 의 병렬이다. 두 가지는 서로 직렬이다 ($Z_T = Z_1 + Z_2$).

(a) $Z_L = j\omega L$, $Z_C = 1/(j\omega C)$ 를 구하라.

(b) 병렬부 $Z_2 = R_2 \parallel Z_C$ 를 직교형($a + jb$)으로 구하라.

(c) 전체 등가 임피던스 Z_T 와 $|Z_T|$, $\angle Z_T$ 를 구하라.

[생성 문제 · 9~15주차 강의노트 범위 · 해답: 생성문제_해답.md]

[생성 9.93] 정현파 회로 — 페이지 전압 분배

$V_s = 120\angle 0^\circ \text{ V(rms)}$, $\omega = 2000 \text{ rad/s}$. 전원에 $R = 40 \, \Omega$ 이 직렬, 그 뒤 $L = 20 \text{ mH}$ 와 $C = 10 \, \mu\text{F}$ 의 병렬($L \parallel C$)이 연결된다. 병렬부 양단 전압 V_o 를 구한다.

(a) Z_L , Z_C , 병렬부 $Z_p = L \parallel C$ 를 구하라.

(b) 전압 분배 $V_o = V_s \cdot Z_p / (R + Z_p)$ 를 크기·위상으로 구하라.

(c) V_o 와 V_s 의 위상차를 말하라.

[생성 문제 · 9~15주차 강의노트 범위 · 해답: 생성문제_해답.md]

10.42 The phasor voltage $\mathbf{V}_{ab}$ in the circuit shown in Fig. P10.42 is $300\angle 0^\circ$ V (rms) when no external load is connected to the terminals a, b. When a load having an impedance of $200 - j500\ \Omega$ is connected across a, b, the value of $\mathbf{V}_{ab}$ is $156 - j42$ V (rms).

Problems **435**

- a) Find the impedance that should be connected across a, b for maximum average power transfer.
- b) Find the maximum average power transferred to the load of (a).
- c) Construct the impedance of part (a) using components from Appendix H if the source frequency is 500 Hz.

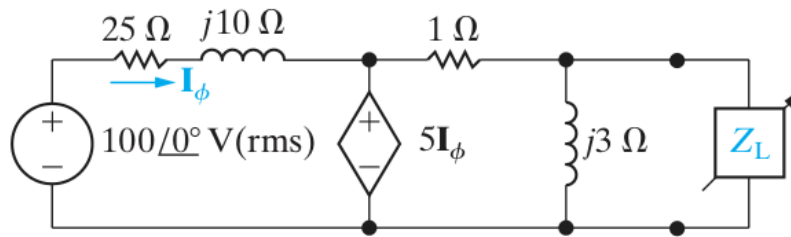
Figure P10.42

— 풀이 —

10.44 The load impedance Z_L for the circuit shown in Fig. P10.44 is adjusted until maximum average power is delivered to Z_L .

- Find the maximum average power delivered to Z_L .
- What percentage of the total power developed in the circuit is delivered to Z_L ?

Figure P10.44

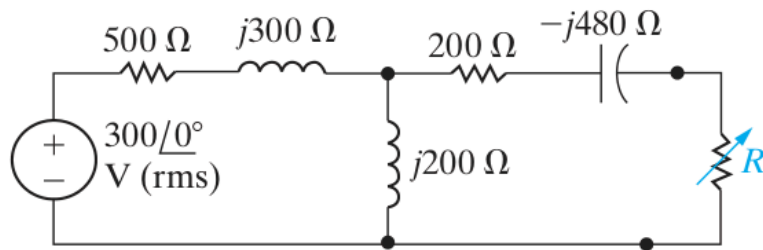


— 풀이 —

10.46 The variable resistor in the circuit shown in Fig. P10.46 is adjusted until the average power it absorbs is maximum.

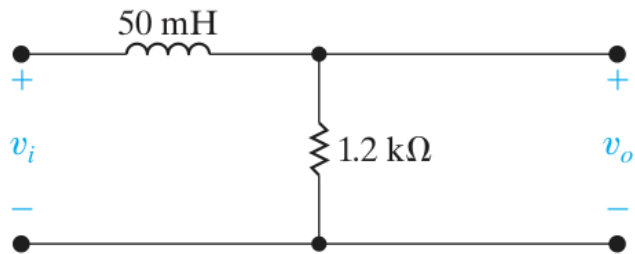
- Find R .
- Find the maximum average power.
- Find a resistor in Appendix H that would have the most average power delivered to it.

Figure P10.46



- 14.1** a) Find the cutoff frequency in hertz for the RL filter shown in Fig. P14.1.
- b) Calculate $H(j\omega)$ at ω_c , $0.125\omega_c$, and $8\omega_c$.
- c) If $v_i = 20 \cos \omega t$ V, write the steady-state expression for v_o when $\omega = \omega_c$, $\omega = 0.125\omega_c$, and $\omega = 8\omega_c$.

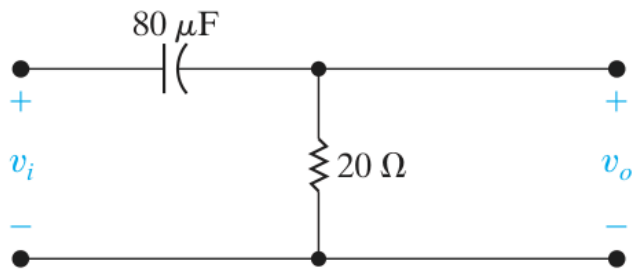
Figure P14.1



— 풀이 —

- 14.11** a) Find the cutoff frequency (in hertz) for the high-pass filter shown in Fig. P14.11.
- b) Find $H(j\omega)$ at ω_c , $0.125\omega_c$, and $8\omega_c$.
- c) If $v_i = 75 \cos \omega t$ V, write the steady-state expression for v_o when $\omega = \omega_c$, $\omega = 0.125\omega_c$, and $\omega = 8\omega_c$.

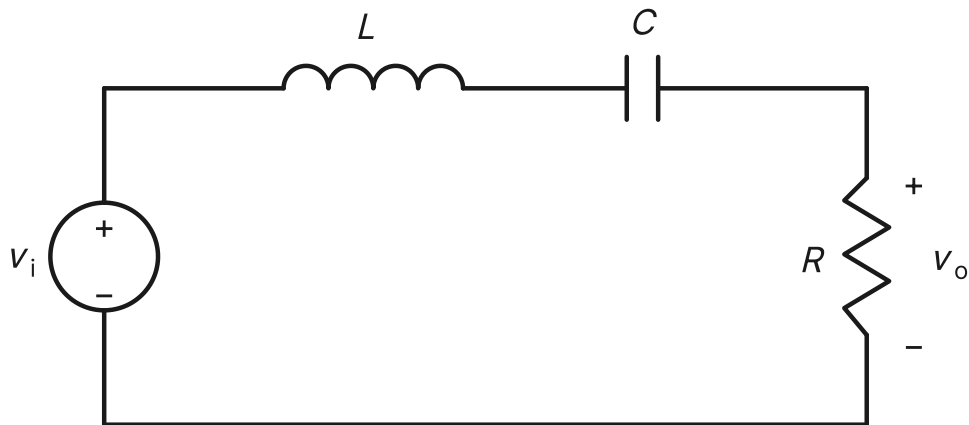
Figure P14.11



14.21 Design a series RLC bandpass filter (see Fig. 14.19[a]) with a quality of 8 and a center frequency of 50 krad/s, using a $0.01\ \mu\text{F}$ capacitor.

- Draw your circuit, labeling the component values and output voltage.
- For the filter in part (a), calculate the bandwidth and the values of the two cutoff frequencies.

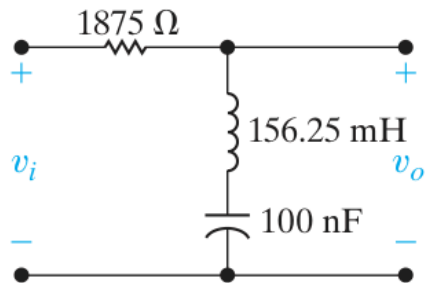
Fig. 14.19[a] — 직렬 RLC 대역통과 필터 (출력 v_o 는 R 양단)



— 풀이 —

- 14.35** For the bandreject filter in Fig. P14.35, calculate
 (a) ω_o ; (b) f_o ; (c) Q ; (d) β in hertz; (e) ω_{c1} ; (f) f_{c1} ;
 (g) ω_{c2} ; and (h) f_{c2} .

Figure P14.35

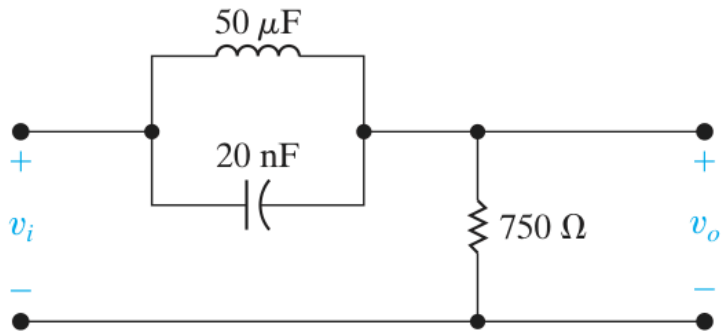


— 풀이 —

14.38 For the bandreject filter in Fig. P14.38, calculate (a) ω_o ; (b) f_o ; (c) Q ; (d) ω_{c1} ; (e) f_{c1} ; (f) ω_{c2} ; (g) f_{c2} ; and (h) β in kilohertz.

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Figure P14.38



— 풀이 —